Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
2	(a)	(i)	$I_1 = I_2 = I_3$ **or** two of $I_1 = I_2$, $I_2 = I_3$, $I_1 = I_3$	1			1		
		(ii)	$V_1 + V_2 = V_3$	1			1		
	(b)	(i)	Correct application of any potential divider equation, e.g. $\frac{V_{\text{out}}}{12\text{ V}} = \frac{32\ \Omega}{48\Omega}$ **or** current = $\frac{12\text{ V}}{48\ \Omega}$ [= 0.25 A] (1) $V_{\text{out}} = 8.0$ [V] if it follows clearly from some correct working (1)		2		2	2	
		(ii)	$V_{\text{out}} = \left[12\text{ V} \times \frac{12\ \Omega}{48\Omega}\right] = 3.0$ [V] **or** [0.25 **ecf** × 12] = 3.0 [V]		1		1	1	
	(c)		R_A and R_B each have 2.0 V across them and the $20\ \Omega$ resistor has 8.0 V across it (1) $R_\text{A} = 5$ [Ω], $R_\text{B} = 5$ [Ω] (1) **Alternative 1 ($V = IR$) award 2 marks for:** $I = \frac{2}{R_B} = \frac{10}{20+R_B}$ leads to $R_B = 5$ [Ω] Then $I = \frac{10}{25} = 0.4$ [A] Then $R_A = \frac{2}{0.4} = 5$ [Ω] **Alternative 2 ($\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R}{R_T}$)** For R_B + 20: $\frac{10}{12} = \frac{R_B+20}{R_A+20+R_B}$ and for R_B: $\frac{2}{12} = \frac{R_B}{R_A+20+R_B}$ (1) Multiplying by 5 gives $\frac{10}{12} = \frac{5R_B}{R_A+20+R_B}$ and so $5R_B = R_B + 20$ $R_B = 5$ [Ω] and with a sub back in $R_A = 5$ [Ω] (1)			2	2		
			Question 2 total	**2**	**3**	**2**	**7**	**3**	**0**